

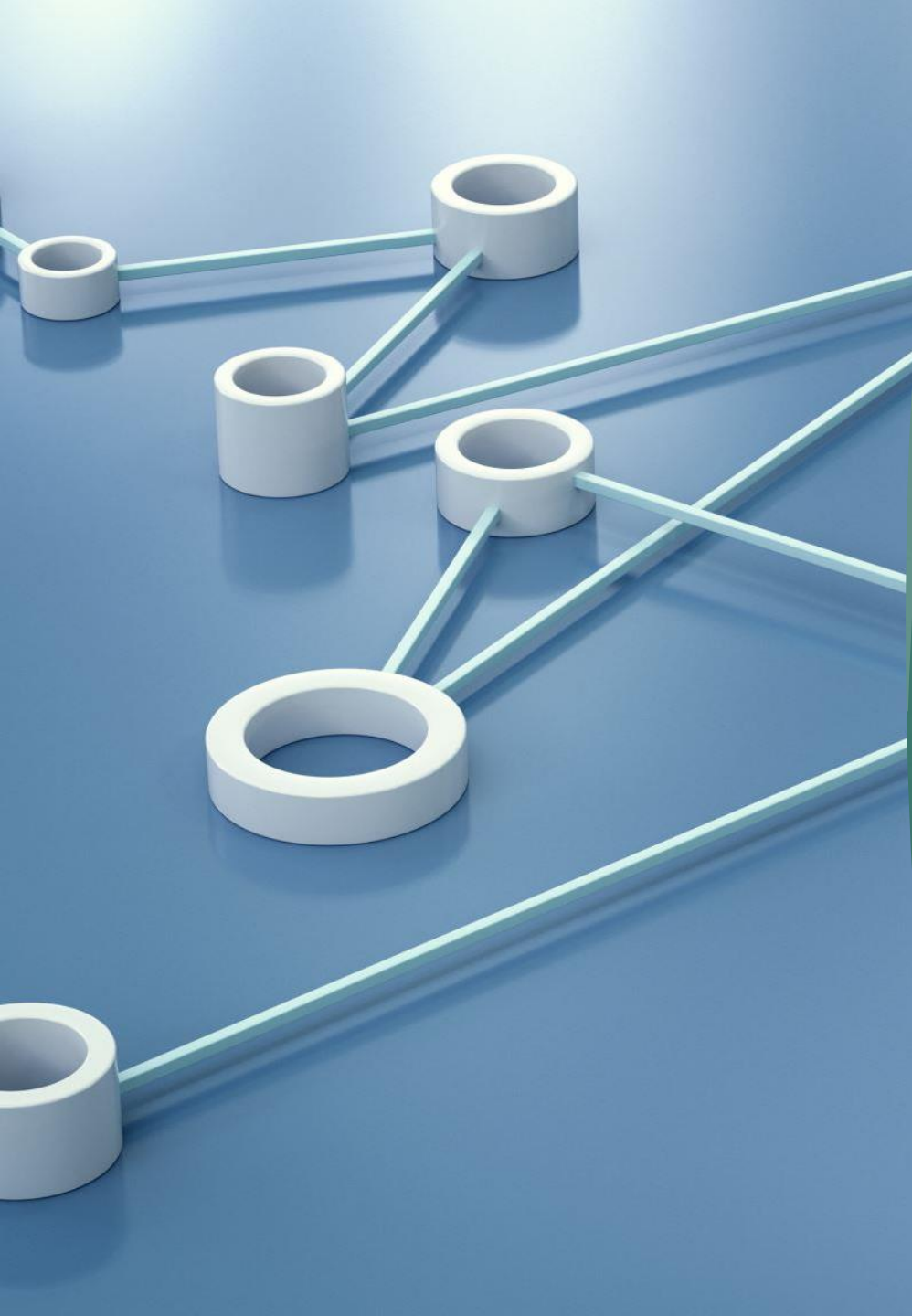


Horizon Health Network Data-Driven Bed Capacity & Demand Planning

Using admissions, occupancy and workforce data to move from reactive to proactive capacity management.

Presented by Oluwatosin Adeniran





INTRODUCTION

 Horizon Health Network is a leading UK private healthcare provider delivering elective and emergency care across a growing network of 37 hospitals.

 Its expansion has increased patient volumes and operational complexity, with hospitals managing multiple specialities, wards, admissions, diagnostics and discharge pathways.

 Supported by digital systems such as electronic health records, admission-discharge-transfer systems and bed-management platforms, the organisation generates extensive operational data.

 However, much of this information remains fragmented, creating an opportunity to transform existing data into clear, consolidated insight that supports better capacity planning, operational efficiency and patient flow.

The Business Problem

Horizon Health Network faces persistent uncertainty in bed availability

Hospitals operate 24/7 across multiple wards and specialities, with demand influenced by:

-  Emergency admissions
-  Elective procedures
-  Patient transfers
-  Discharge delays
-  Speciality-specific care pathways
-  Seasonal fluctuations

The core problem : Planners lack a consolidated view of where and when bed pressure is building.

Fluctuating demand → Uncertain bed availability → Reactive decisions → Patient & financial impact

Fluctuating patient demand makes bed availability difficult to predict, causing hospitals to respond reactively, which increases operational costs and negatively affects patient flow and experience.

The Analytical Solution

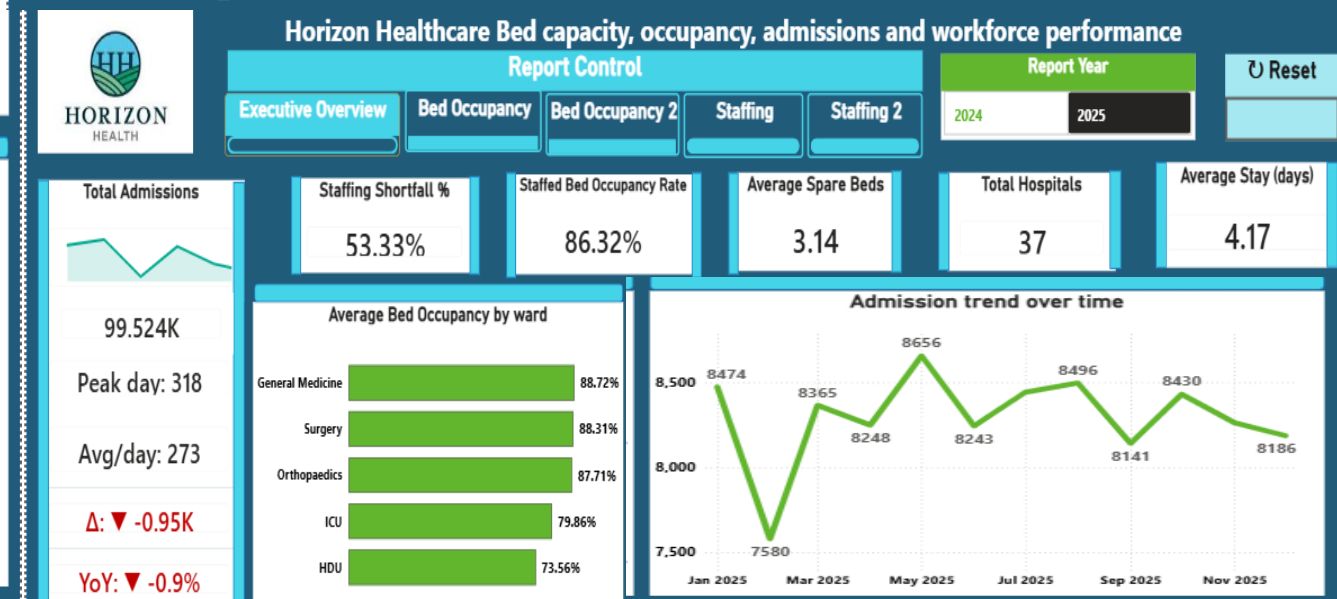
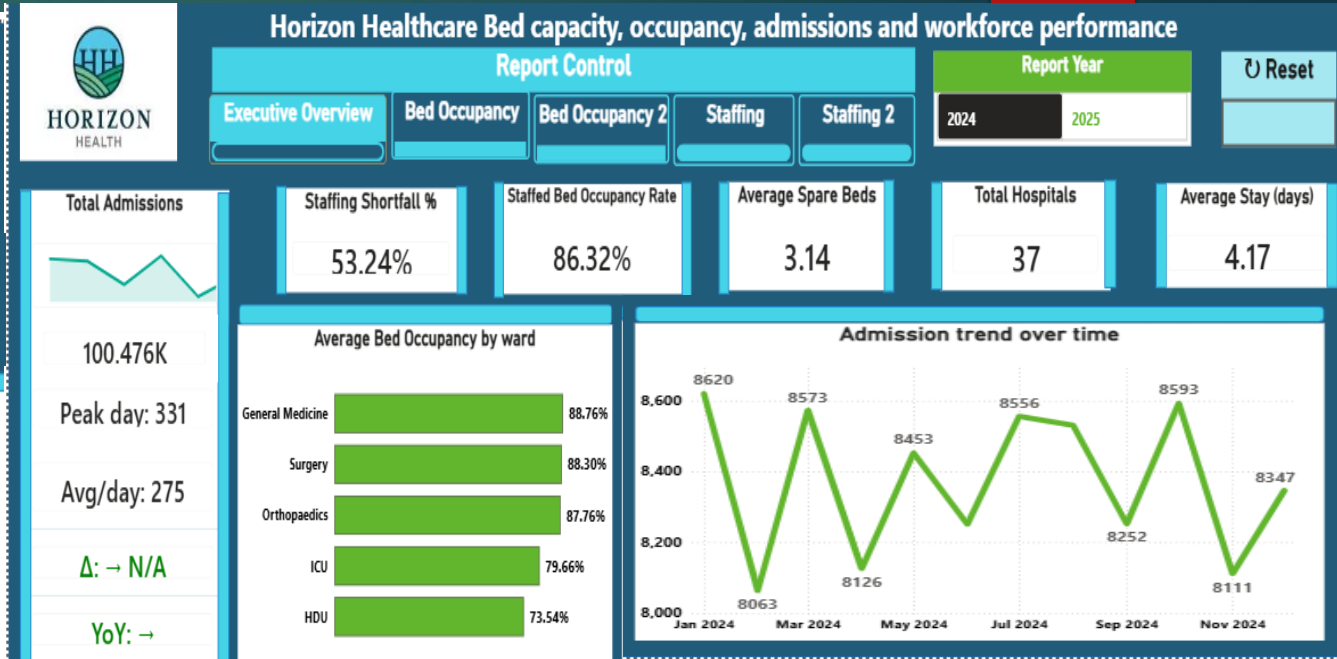
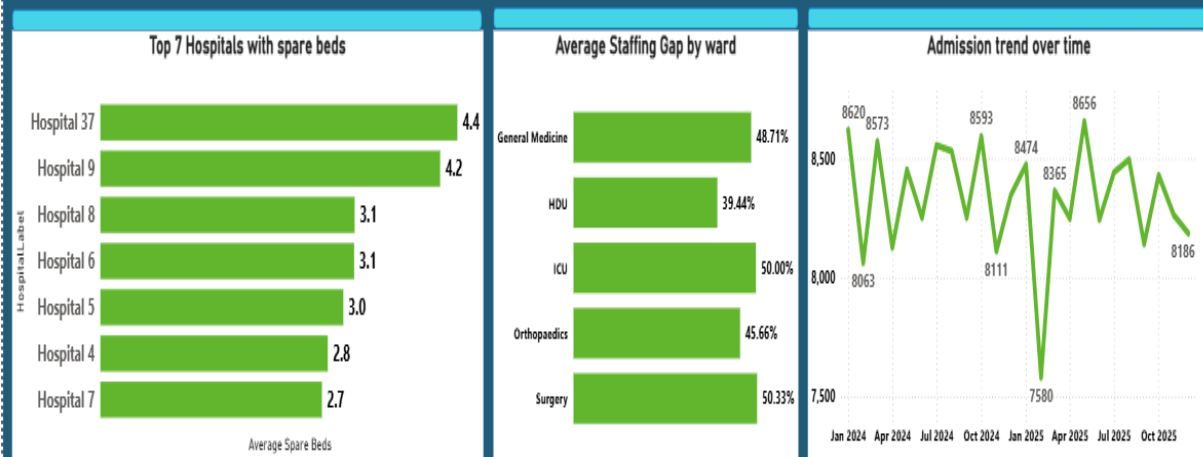
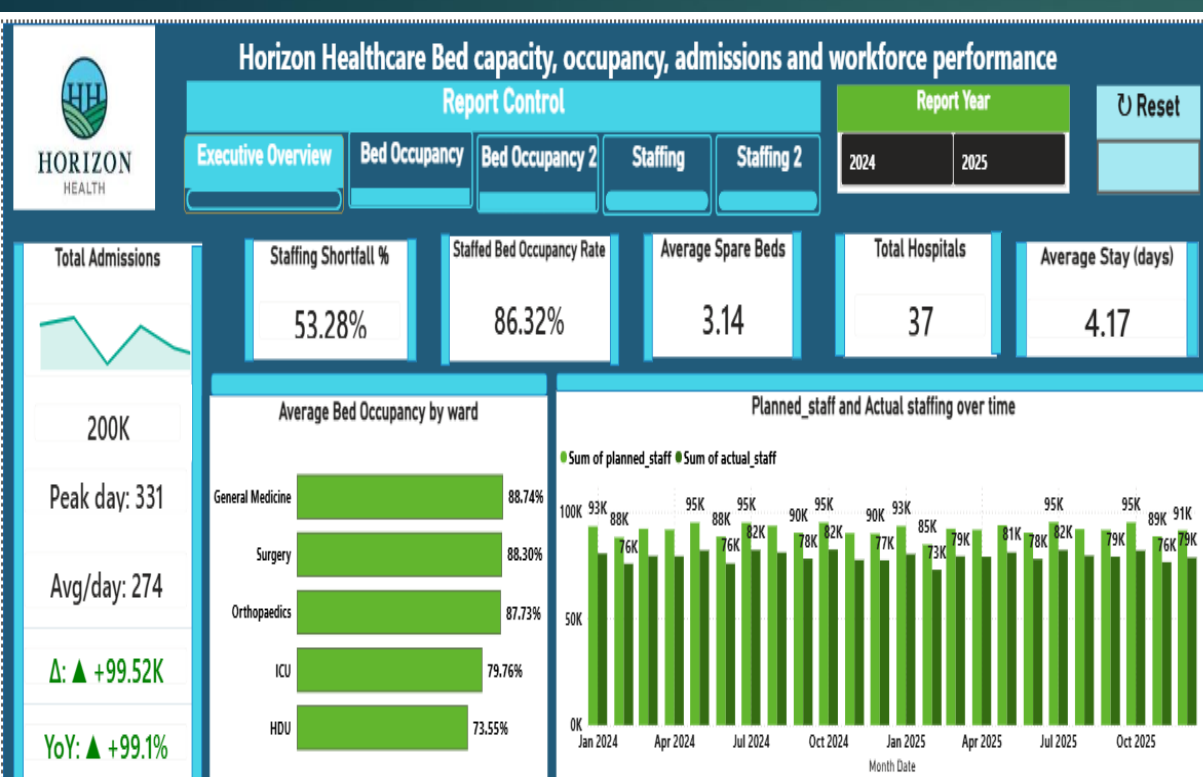
A consolidated Power BI capacity-planning dashboard

Admissions → Occupancy → Bed Pressure → Staffing

The dashboard provides:

-  Interactive time-period filtering
-  Admissions trends
-  Year-over-year comparison
-  Delta analysis
-  Staffed bed occupancy
-  Bed pressure monitoring
-  Staffing gap analysis
-  Historical trend visibility

Executive Overview



BED OCCUPANCY AND DEMAND



HORIZON
HEALTH

Horizon Healthcare Bed capacity, occupancy, admissions and workforce performance

Report Control

Executive Overview

Bed Occupancy

Bed Occupancy 2

Staffing

Staffing 2

Report Year

2024

2025

Reset

Average Beds



23.65

Average Occupied beds



19.82

Average Closed Beds



69.35%

Bed Pressure



83.79%

Peak Occupancy

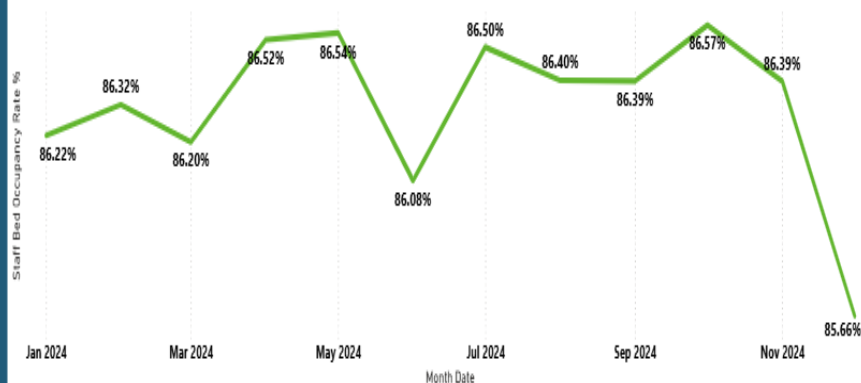


88.98%

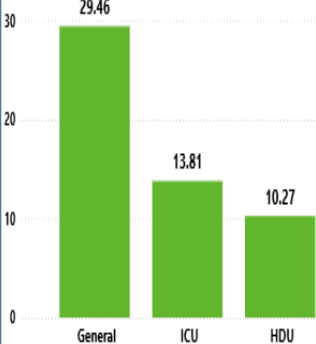
Average Occupancy by bed type



Bed Occupancy trend over time



Average bed capacity by bed types



Occupied beds and Available Beds by Month Date



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23.65

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19.82

Average Closed Beds



69.05%

Bed Pressure



83.80%

Peak Occupancy

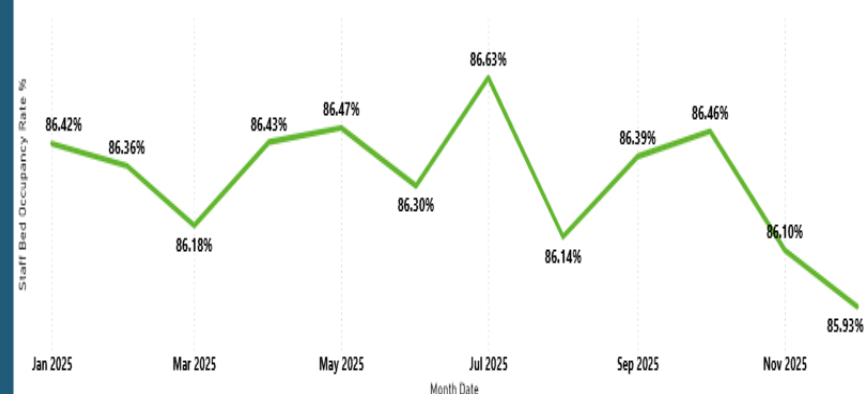


88.74%

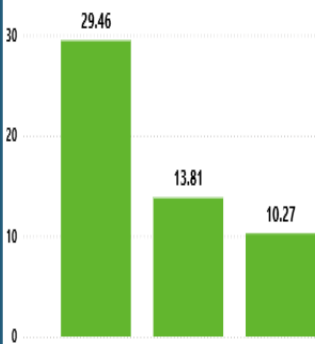
Average Occupancy by bed type



Bed Occupancy trend over time



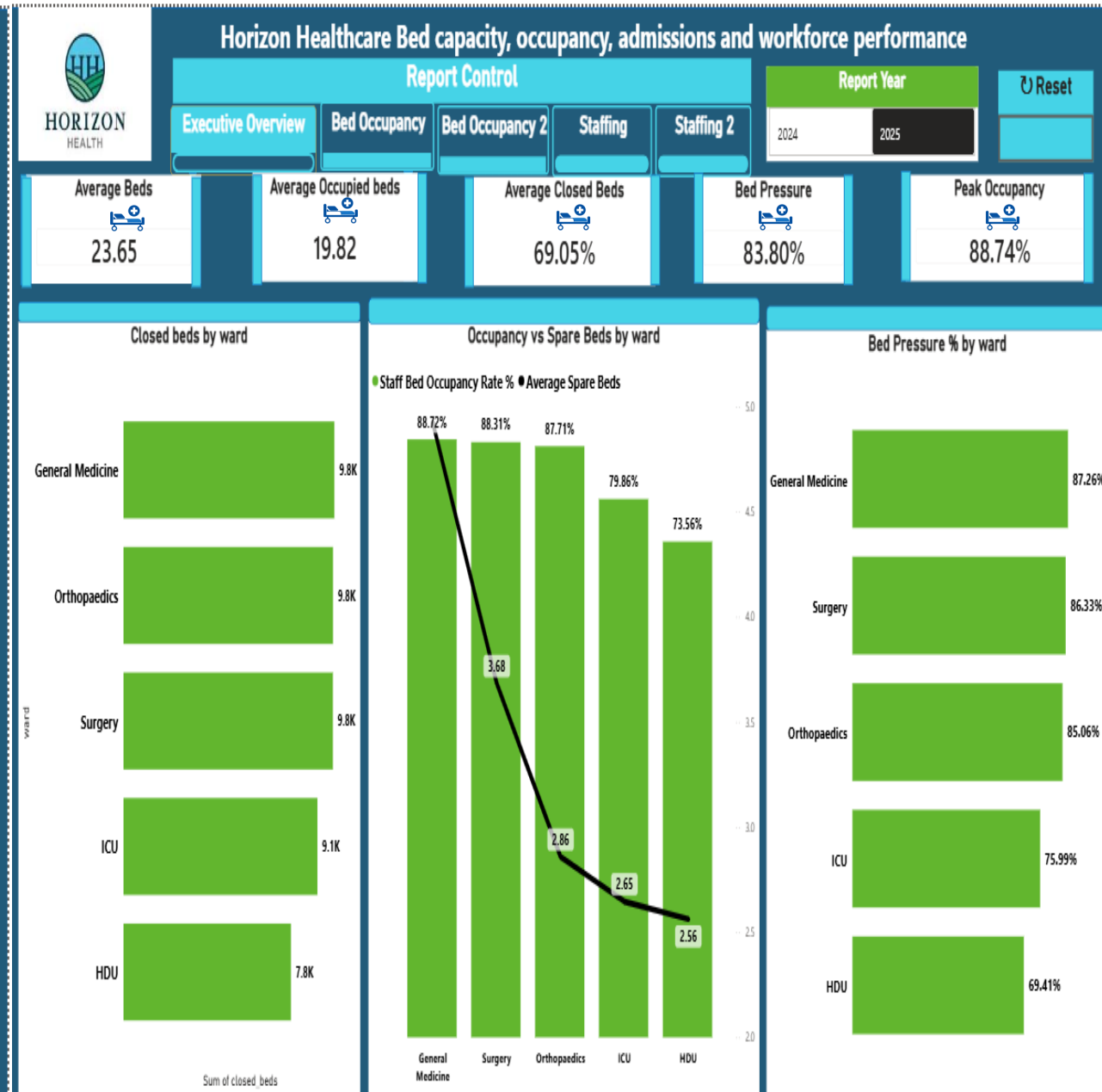
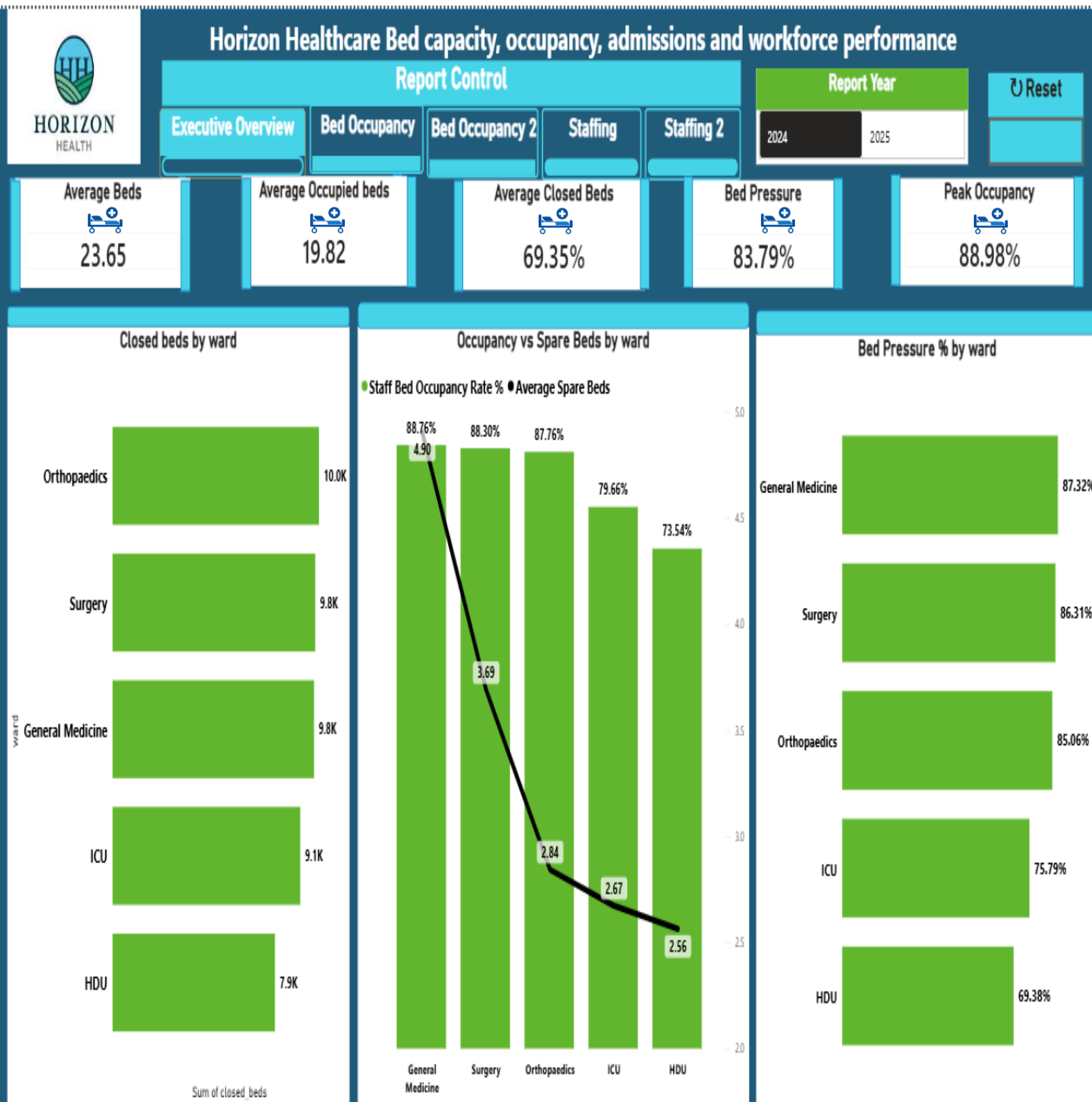
Average bed capacity by bed types



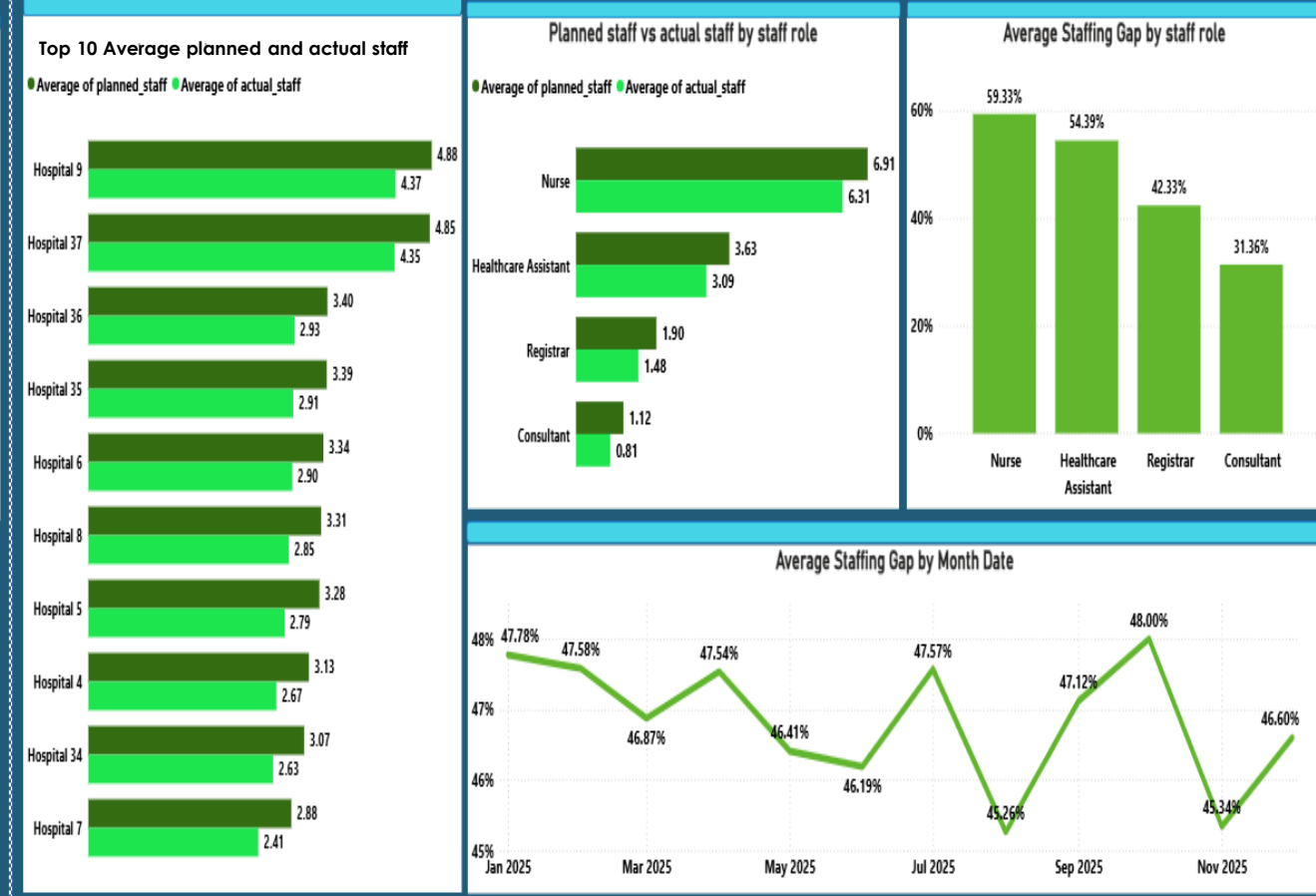
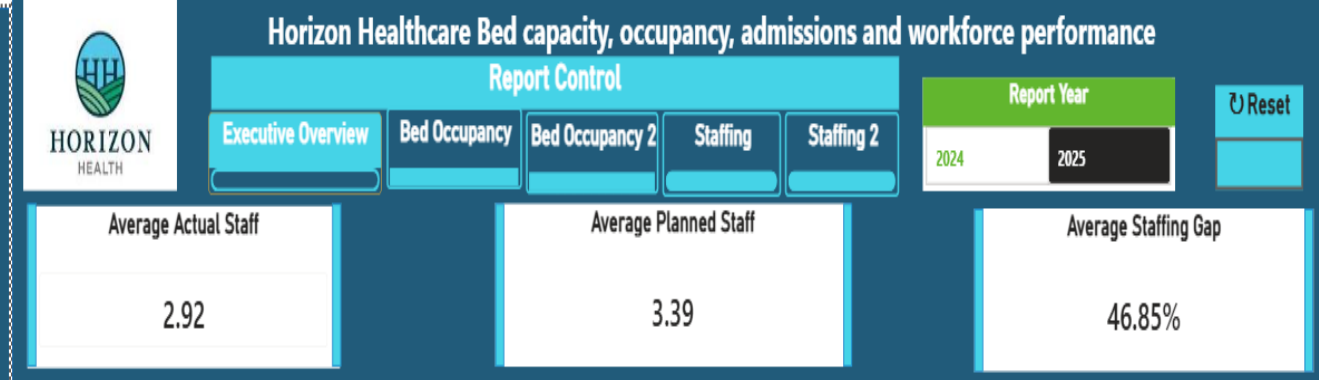
Occupied beds and Available Beds by Month Date



BED OCCUPANCY AND DEMAND 2



STAFFING



STAFFING



HORIZON
HEALTH

Horizon Healthcare Bed capacity, occupancy, admissions and workforce performance

Report Control

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Bed Occupancy

Bed Occupancy 2

Staffing

Staffing 2

Report Year

2024

2025

Reset

Average Actual Staff

2.92

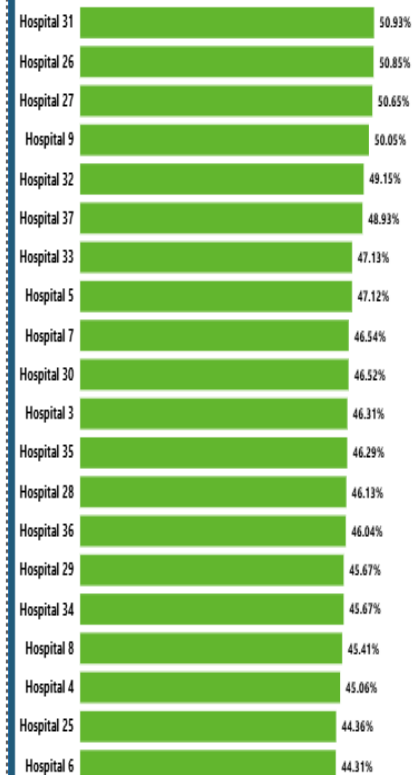
Average Planned Staff

3.39

Average Staffing Gap

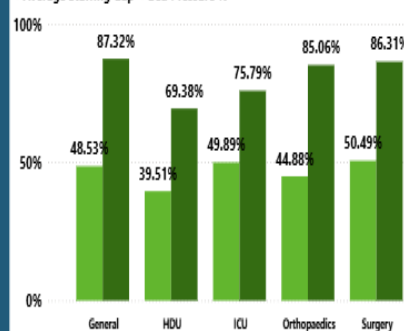
46.52%

Top 20 Average Staffing Gap by Hospital

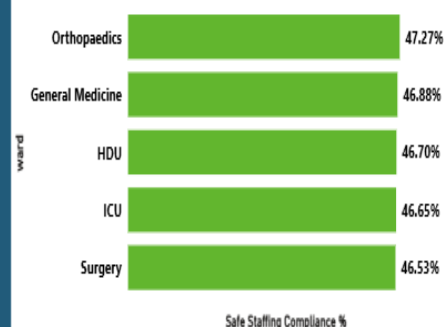


Staffing Gap and Bed Pressure by ward

● Average Staffing Gap ● Bed Pressure %



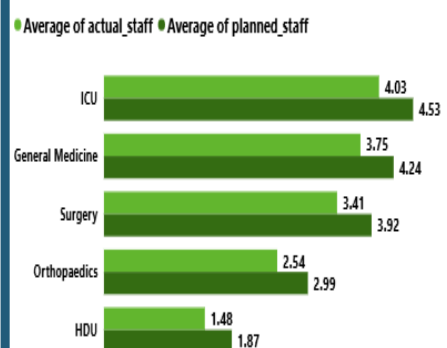
Safe Staffing Compliance % by ward



Average Staffing Gap by ward and role

ward	Consultant	Healthcare Assistant	Nurse	Registrar	Total
General Medicine	30.51%	58.74%	60.60%	44.27%	48.53%
HDU	29.28%	43.97%	52.72%	32.09%	39.51%
ICU	36.72%	57.58%	58.21%	47.06%	49.89%
Orthopaedics	30.04%	52.18%	55.76%	41.56%	44.88%
Surgery	34.22%	58.65%	60.43%	48.66%	50.49%
Total	32.15%	54.22%	57.54%	42.73%	46.66%

Average actual staff and planned staff by ward



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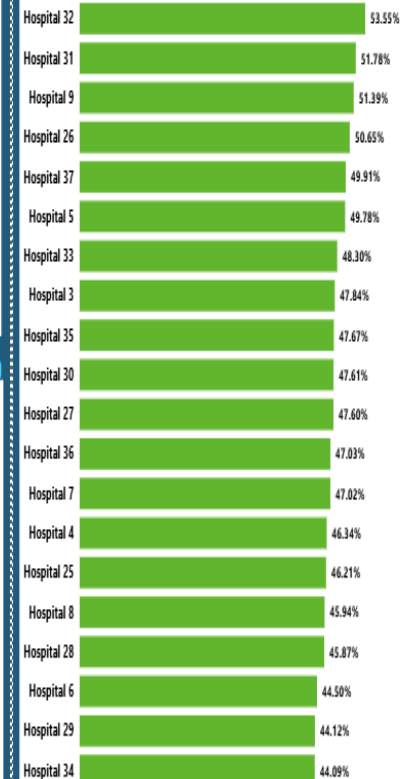
Average Planned Staff

3.39

Average Staffing Gap

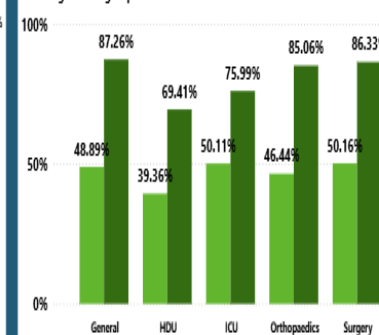
46.85%

Top 20 Average Staffing Gap by Hospital

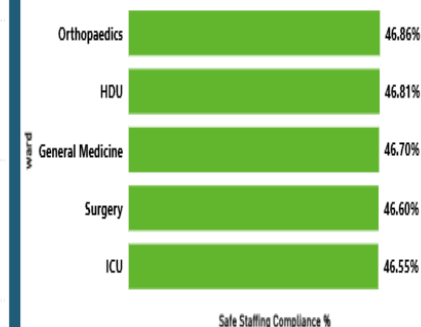


Staffing Gap and Bed Pressure by ward

● Average Staffing Gap ● Bed Pressure %



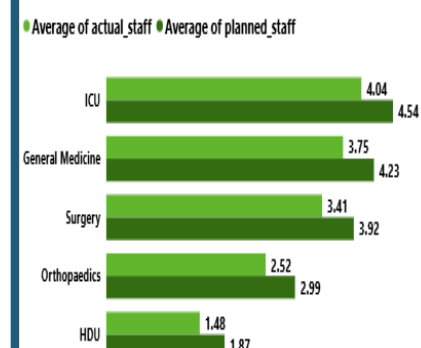
Safe Staffing Compliance % by ward



Average Staffing Gap by ward and role

ward	Consultant	Healthcare Assistant	Nurse	Registrar	Total
General Medicine	31.06%	58.67%	60.13%	45.68%	48.89%
HDU	29.17%	43.29%	53.68%	31.29%	39.36%
ICU	35.99%	58.65%	60.03%	45.77%	50.11%
Orthopaedics	28.76%	53.59%	59.70%	43.73%	46.44%
Surgery	33.76%	57.81%	61.69%	47.40%	50.16%
Total	31.75%	54.40%	59.04%	42.77%	46.99%

Average actual staff and planned staff by ward



Proposed Business Solution

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1. Consolidate operational data

- ▶ Create a unified analytical layer combining:
- ▶ Admissions
- ▶ Bed occupancy
- ▶ Staffed beds
- ▶ Staffing
- ▶ Elective activity
- ▶ Discharge information

2. Establish early-warning indicators

Monitor:

Occupancy rate

Bed pressure

Admissions trend

Staffing gap

High-demand periods

3. Introduce forward-looking planning

Using historical patterns to identify:

- Seasonal demand
- Recurring pressure periods
- Specialty/ward bottlenecks
- Capacity requirements

Recommendation to Management

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Now- Improve visibility

- Use the dashboard as a central operational view
- Monitor occupancy and pressure routinely
- Identify emerging high-pressure periods earlier

Next - Improve coordination

- Coordinate bed allocation with admissions and elective scheduling
- Strengthen discharge planning
- Align staffing with expected demand
- Introduce escalation triggers

Longer Term -Increase effective capacity

- Review staffed-bed requirements
- Identify persistent specialty bottlenecks
- Evaluate additional capacity where sustained pressure exists
- Improve workforce planning

Business Benefits

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Current state

Reactive bed management
Fragmented data
Manual reporting
Late identification of pressure
Last-minute reallocations
Overtime/agency pressure
Patient delays

Future state

Proactive capacity planning
Consolidated view
Repeatable dashboard
Early warning
Planned interventions
Better workforce alignment
Improved flow

Conclusion

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- ▶ **Three key management insights**

- 1. Demand is relatively stable**

- ▶ Admissions did not materially increase between 2024 and 2025.

- 2. Capacity remains under pressure**

- ▶ High utilisation of staffed beds reduces the available buffer.

- 3. The solution is proactive capacity management**

- ▶ Horizon needs a consolidated view of demand, occupancy and workforce capacity to identify pressure before it becomes critical.



THANK
YOU